Evaluating how well AI can understand spoken conversations about data visualizations

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1. The Core Challenge

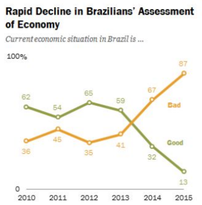
For an AI to be a useful meeting assistant, it must deeply understand the content of the conversation. This includes not just the words spoken, but the context—the data visualizations being discussed.

***The Problem:** How can AI reliably connect a spoken phrase to the specific elements of a chart?*

2. The Gap

Chart Question Answering (CQA): Design systems that answer questions based on information presented in charts and visualizations, requiring the analysis of both visual and textual components within the chart.

The Gap: These benchmarks test chart interpretation in a vacuum, not an AI's ability to interpret a conversation about the chart between people



Q1: Which year has the most divergent opinions about Brazil's economy?
Answer: 2015

Q2: What is the peak value of the orange line?
Answer: 87

Example taken from the ChartQA test

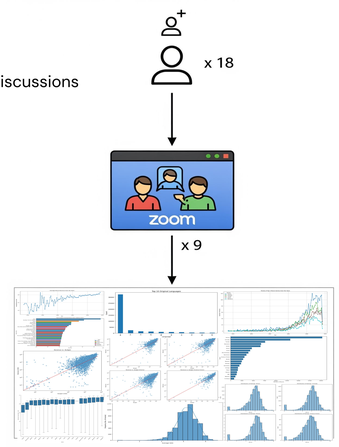
The **ChartQA** assessment test shows that models can answer factual questions about a chart.

3. Design Strategy

1. Conversational Corpus: 72 transcript-chart pairs from discussions mimicking online meetings.

Session Name	Total Word Count	Duration (in minutes)
P5_P6	3945	34
P7_P8	4133	35
P9_P10	7343	55
P11_P12	5611	52
P13_P14	6951	53
P15_P16	4605	26
P17_P18	4893	32
P19_P20	2417	17
P21_P22	2228	17

	Pt 1	Pt 2	Pt 3	Pt 4	Pt 5	Pt 6	Pt 7	Pt 8
P5_P6	590	376	407	366	378	429	257	485
P7_P8	534	447	655	497	593	462	181	224
P9_P10	763	591	712	1021	1270	1227	375	871
P11_P12	1354	689	363	610	447	729	428	387
P13_P14	712	410	394	688	793	926	1846	473
P15_P16	961	676	313	437	499	329	592	343
P17_P18	562	752	582	683	366	484	612	332
P19_P20	257	204	264	225	269	283	234	161
P21_P22	340	482	287	84	164	88	150	143



Discussing 8 Charts in each session

2. Benchmark: 318 questions based on real life data visualization discussions

3. Dual Axis Framework: Labeling questions inspired by Bloom's Taxonomy on two axes.

AXIS 1 Labels (Primary Category)	Bloom's Comprehension Levels
Factual Recall	Level 1: "Remembering"
Data Interpretation	Level 2: "Understanding"
Participant State Analysis	Level 2: "Applying"
Causal & Process Analysis	Level 3: "Analyzing"

AXIS 2 Labels (Question Topic Tags)	
1. Chart Layout	5. Data Provenance
2. Axis & Labels	6. Visual Encoding
3. Data Pattern	7. Participant Hypothesis
4. Data Point	

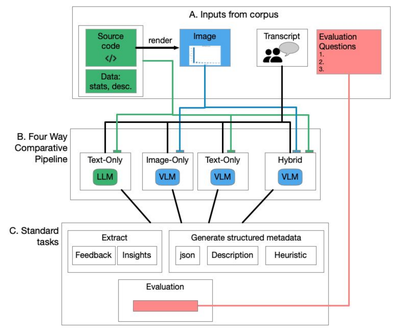
4. Experimental Setup: The 4-Way Pipeline

Pipeline 1 - Text-Only (LLM): Analyzes only the Python source code, data schema, stats, and it's own generated prose, heuristic & JSON spec of the code. It does not see the actual generated chart images.

Pipeline 2 - Image-Only (VLM): Analyzes only the rendered image of the chart, and it's own generated prose, heuristic & JSON spec of the image. Tests pure visual interpretation. It does not use the chart's source code, the data schema, or the statistical summary of the data.

Pipeline 3 - Text-Only (VLM): Same text-only input as Pipeline 2, but with a VLM. Isolates architectural differences.

Pipeline 4 - Hybrid (VLM): Analyzes both the chart image and the full textual context. Tests multimodal fusion.



High-level flow for pipeline operations: (A) Inputs (B) Four Pipeline Modes (C) Standardized Comprehension Tasks

Prose Description Prompt
You are given {num_items} data visualizations. Describe each visualization in a detailed paragraph. Provide a holistic summary at the end.

Heuristic Description Prompt
Analyze the provided chart(s). Provide a single-sentence heuristic description summarizing all the charts

Specification Prompt
Generate a JSON formatted specification for each of the charts provided.

Feedback Prompt
You are an AI assistant analyzing user feedback on a set of chart/s. Use all the provided context to perform your task.

sources: description, heuristic, chart spec, data schema, statistical summary

Based on ALL of the above context, extract the unique critiques and suggestions from the user transcript. List them as a bulleted list.

Question answering prompt
You are an expert AI assistant. Your mission is to answer the user's question by synthesizing information from ALL the provided sources.

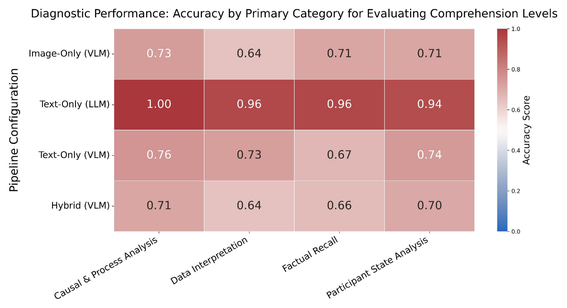
Sources:
Transcript
Feedback
Prose description
Heuristic description
Specification
Question

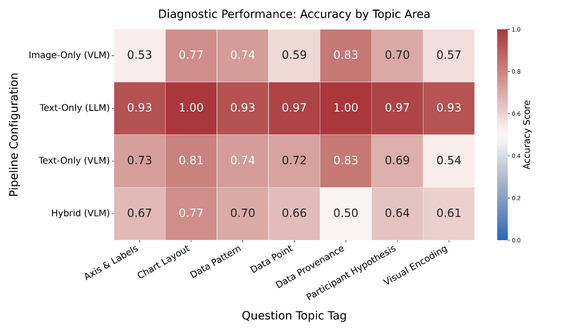
First, choose the single best option (A, B, C, or D). Then, provide a detailed rationale for your choice.
Your response MUST follow this format exactly:
Answer: [Your chosen letter]
Rationale: [Your detailed explanation]

Final Prompts

5. Results

- Text-Only (LLM) is the clear winner
- Other pipelines indicating weaker performance, especially in 'Data Interpretation' and 'Factual Recall' for the VLM-based models.
- Image-Only (VLM) struggling with 'Axis & Labels' (53%).
- Hybrid (VLM) weak on 'Data Provenance' (50%).





6. Findings and Discussions

The Reliability of Textual Ground Truth:

- The Text-Only LLM's success suggests that unambiguous code precisely defines data, mappings, and labels, eliminating misinterpretation.
- It supports the hypothesis that structured text is the more reliable input modality in this case.

The Brittleness of Visual Interpretation:

- The Image-Only pipeline was unreliable, frequently failing on tasks requiring precise data extraction (e.g., reading axis labels or specific data points).
- This supports the "Brittle Vision" hypothesis that vision models may grasp the general layout but cannot be trusted for detailed factual analysis from an image alone.

Knowledge Conflict in Hybrid Modalities:

- The Hybrid pipeline was the worst performer. It got "confused" when visual appearance was ambiguous or different from the provided code.
- Stark Example: The Hybrid VLM scored 0.00 on Data Interpretation for Data Provenance questions, while the Text-Only VLM scored 1.00 on the exact same task. The model trusted its flawed visual perception over the correct ground-truth text.

The Impact of Model Architecture:

- The Text-Only LLM (95.9%) significantly outperformed the Text-Only VLM (72.4%) on the exact same input.
- This shows architecture matters.

Q. Can AI agents understand spoken collaborative discussions about data visualizations in meetings?
A. Yes. We just witnessed LLM performing at 96% accuracy, and even the worst performance (68%) was significantly greater than 50%

Q. What is the most effective way to provide context to an AI agent so it can reliably understand a spoken, collaborative discussion about a data visualization?
A. The structured data happens to be the most effective way to provide context to an AI agent in these cases of evaluations.

7. Implications & Future Directions

Current meeting tools like PowerPoint & Zoom limit AI agents by only providing visualizations as images. We found this image-only data insufficient for models to fully grasp meeting discussions. To fix this, platforms must begin embedding visualization code and text metadata to unlock effective AI support. Future directions:

1. **Expanded Evaluation:** Add more questions to the benchmark for better consistency.
2. **Bias Reduction:** Continue to grow the benchmark and collaborate with other researchers to review and label questions to eliminate bias.
3. **Corpus Expansion:** Conduct more studies to expand the conversational corpus
4. **Advanced Models:** Test more sophisticated models
5. **Latitudinal Analysis:** Investigate techniques for fusing multimodal inputs.
6. **Qualitative Analysis:** Invite study participants back to evaluate the quality of the AI's feedback extraction and suggestions.

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Can AI agents understand spoken conversations about data visualizations in online meetings?

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